

[1]: # Q1. Student Marks Manager

```
marks = [78, 85, 67, 90, 72]
```

```
marks.append(88)
```

```
marks.insert(2, 95)
```

```
marks.remove(67)
```

```
marks.sort()
```

```
print("Updated Marks List:", marks)
```

Updated Marks List: [72, 78, 85, 88, 90, 95]

[2]: # Q2. Tuple Operations

```
cities = ("Delhi", "Mumbai", "Chennai", "Bangalore", "Kolkata", "Hyderabad")
```

```
print("First City:", cities[0])
```

```
print("Last City:", cities[-1])
```

```
print("Total Elements:", len(cities))
```

```
if "Delhi" in cities:
```

```
    print("Delhi exists")
```

```
else:
```

```
    print("Delhi does not exist")
```

First City: Delhi

Last City: Hyderabad

Total Elements: 6

Delhi exists

[3]: # Q3. Set Challenge

```
A = {1, 2, 3, 4, 5}
```

```
B = {4, 5, 6, 7}
```

```
print("Union:", A.union(B))
```

```
print("Intersection:", A.intersection(B))
```

```
print("Difference:", A.difference(B))
```

```
print("Symmetric Difference:", A.symmetric_difference(B))
```

Union: {1, 2, 3, 4, 5, 6, 7}

Intersection: {4, 5}

Difference: {1, 2, 3}

Symmetric Difference: {1, 2, 3, 6, 7}

```
[4]: # Q16. Library Book System
```

```
books = []

def add_book(book):
    books.append(book)

def remove_book(book):
    books.remove(book)

def search_book(book):
    if book in books:
        print("Book Found")
    else:
        print("Book Not Found")

def display_books():
    print("Books:", books)

add_book("Python")
add_book("AI")
search_book("Python")
remove_book("AI")
display_books()
```

```
Book Found
Books: ['Python']
```

```
[5]: # Q17. Student Result Analyzer
```

```
marks = []

for i in range(5):
    mark = int(input("Enter mark: "))
    marks.append(mark)

def average_marks(marks):
    return sum(marks) / len(marks)

def highest_marks(marks):
    return max(marks)

def lowest_marks(marks):
    return min(marks)

def passed_students(marks):
    count = 0
    for mark in marks:
        if mark > 40:
            count += 1
    return count

print("Average:", average_marks(marks))
print("Highest:", highest_marks(marks))
print("Lowest:", lowest_marks(marks))
print("Passed Students:", passed_students(marks))
```

```
Enter mark: 50
Enter mark: 60
Enter mark: 30
Enter mark: 80
Enter mark: 100
Average: 64.0
Highest: 100
Lowest: 30
Passed Students: 4
```

```
[6]: # Q18. Password Strength Checker
```

```
def check_password(password):  
  
    if (len(password) >= 8 and  
        any(char.isdigit() for char in password) and  
        any(char.isupper() for char in password) and  
        any(char.islower() for char in password)):  
  
        return "Strong Password"  
  
    else:  
        return "Weak Password"  
  
password = input("Enter password: ")  
print(check_password(password))
```

```
Enter password: chutti  
Weak Password
```

```
[7]: # Q19. Contact Book Application
```

```
contacts = {}  
  
def add_contact(name, number):  
    contacts[name] = number  
  
def search_contact(name):  
    if name in contacts:  
        print(name, ":", contacts[name])  
    else:  
        print("Contact not found")  
  
def delete_contact(name):  
    if name in contacts:  
        del contacts[name]  
  
def display_contacts():  
    print(contacts)  
  
add_contact("Ram", 9876543210)  
search_contact("Ram")  
display_contacts()  
delete_contact("Ram")  
display_contacts()
```

```
Ram : 9876543210  
{'Ram': 9876543210}  
{}
```

```
[1]: # Q23. Unique Number Counter

numbers = []

for i in range(10):
    num = int(input("Enter number: "))
    numbers.append(num)

unique_numbers = set(numbers)

print("Unique Numbers Count:", len(unique_numbers))

Enter number: 12
Enter number: 24
Enter number: 36
Enter number: 48
Enter number: 52
Enter number: 68
Enter number: 72
Enter number: 84
Enter number: 91
Enter number: 108
Unique Numbers Count: 10
```

```
[2]: # Q24. Voting Eligibility Checker

ages = (12, 18, 25, 16, 40)

def check_voting(age):
    if age >= 18:
        return "Eligible"
    else:
        return "Not Eligible"

for age in ages:
    print(age, "-", check_voting(age))

12 - Not Eligible
18 - Eligible
25 - Eligible
16 - Not Eligible
40 - Eligible
```

```
[3]: # Q25. Word Frequency Counter

sentence = input("Enter sentence: ")

words = sentence.split()

frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

print(frequency)

Enter sentence: Hello world
{'Hello': 1, 'world': 1}
```

```
[8]: # Q20. Mini Banking System

balance = 1000

def deposit(amount):
    global balance
    balance += amount
    print("Balance after deposit:", balance)

def withdraw(amount):
    global balance

    if amount <= balance:
        balance -= amount
        print("Balance after withdrawal:", balance)
    else:
        print("Insufficient Balance")

def check_balance():
    print("Current Balance:", balance)

deposit(500)
withdraw(300)
check_balance()

Balance after deposit: 1500
Balance after withdrawal: 1200
Current Balance: 1200
```

```
[9]: # Q21. HackerRank Style Problem

numbers = [10, 20, 30, 40, 50]

# Find Maximum
maximum = numbers[0]

for num in numbers:
    if num > maximum:
        maximum = num

print("Maximum:", maximum)

# Find Minimum
minimum = numbers[0]

for num in numbers:
    if num < minimum:
        minimum = num

print("Minimum:", minimum)

# Reverse List
reversed_list = []

for i in range(len(numbers)-1, -1, -1):
    reversed_list.append(numbers[i])

print("Reversed List:", reversed_list)

Maximum: 50
Minimum: 10
Reversed List: [50, 40, 30, 20, 10]
```

```
[15]: # Q11. Attendance System

students = []

def add_student(name):
    students.append(name)

def remove_student(name):
    students.remove(name)

def display_students():
    print("Students:", students)

add_student("Rahul")
add_student("Anu")
remove_student("Rahul")
display_students()

Students: ['Anu']
```

```
[16]: # Q12. Employee Salary Dictionary

employees = {}

def add_employee(name, salary):
    employees[name] = salary

def update_salary(name, salary):
    employees[name] = salary

def search_employee(name):
    if name in employees:
        print(name, "Salary:", employees[name])
    else:
        print("Employee not found")

add_employee("Ravi", 30000)
update_salary("Ravi", 35000)
search_employee("Ravi")

Ravi Salary: 35000
```

[12]: # Q8. Largest Among Three Numbers

```
def largest(a, b, c):  
    return max(a, b, c)  
  
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
c = int(input("Enter third number: "))  
  
print("Largest Number:", largest(a, b, c))
```

Enter first number: 10
Enter second number: 7
Enter third number: 20
Largest Number: 20

[13]: # Q9. Factorial using Function

```
def factorial(n):  
    fact = 1  
    for i in range(1, n + 1):  
        fact *= i  
    return fact  
  
num = int(input("Enter number: "))  
print("Factorial:", factorial(num))
```

Enter number: 4
Factorial: 24

[14]: # Q10. Prime Number Checker

```
def is_prime(n):  
    if n <= 1:  
        return False  
  
    for i in range(2, n):  
        if n % i == 0:  
            return False  
  
    return True  
  
num = int(input("Enter number: "))  
  
if is_prime(num):  
    print("Prime Number")  
else:  
    print("Not Prime")
```

Enter number: 10
Not Prime

[4]: # Q4. Dictionary Student Record

```
student = {  
    "name": "Rahul",  
    "roll": 101,  
    "branch": "CSE",  
    "marks": 85  
}  
  
student["marks"] = 92  
  
print("Keys:", student.keys())  
print("Values:", student.values())
```

Keys: dict_keys(['name', 'roll', 'branch', 'marks'])
Values: dict_values(['Rahul', 101, 'CSE', 92])

[5]: # Q5. Mini Shopping Cart

```
items = ["Pen", "Book", "Bag", "Bottle"]  
prices = [10, 50, 500, 100]  
  
total_bill = sum(prices)  
  
print("Total Bill:", total_bill)  
print("Highest Price:", max(prices))  
print("Lowest Price:", min(prices))
```

Total Bill: 660
Highest Price: 500
Lowest Price: 10

[11]: # Q6. Simple Calculator Function

```
def addition(a, b):  
    return a + b  
  
def subtraction(a, b):  
    return a - b  
  
def multiplication(a, b):  
    return a * b  
  
def division(a, b):  
    return a / b  
  
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
  
print("Addition:", addition(a, b))  
print("Subtraction:", subtraction(a, b))  
print("Multiplication:", multiplication(a, b))  
print("Division:", division(a, b))
```

Enter first number: 1